

# PANKHURI KULSHRESTHA

*Machine Learning Research Engineer*

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## PROFILE

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Machine Learning Research Engineer with 8+ years of experience building AI systems for industrial inspection, federated learning, and computer vision. Currently at Mindtrace (Manchester), building distributed monitoring agents for weld inspection workflows and leading 3D deep learning research. Published at IEEE and Springer; work under review at IEEE Transactions on Neural Networks and Learning Systems.

## EXPERIENCE

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### Mindtrace — Machine Learning Research Engineer

Jan 2023 – Present

*Manchester, UK*

- Designing the *agents* package for the Mindtrace open-source framework — autonomous monitoring agents for distributed weld inspection workflows, coordinating tasks and maintaining memory across sessions.
- Designed monorepo architecture and built the *core* package: config injection, structured logging, and observable state primitives.
- Self-supervised learning with DINO architecture features on large-scale unlabelled datasets; parameter-efficient fine-tuning via LoRA achieving **95% accuracy** on industrial inspection tasks.
- 3D deep learning for point cloud segmentation: trained and benchmarked PointNet, PointNet++, and SPVCNN on LiDAR-captured infrastructure data.
- Built LiDAR preprocessing pipelines (PDAL, GeoPandas, Shapely) for coordinate transformations, ground filtering, and tiling at scale.
- Deployed models on AWS, GCP, and Azure; MLOps infrastructure with Docker, MLFlow, and FastAPI microservices (92% detection accuracy).

### Edcast — Computer Vision Engineer

Jun 2021 – Dec 2021

*India*

- Deployed YOLOv3 on the Kneron KL520 USB AI dongle — model quantisation, ONNX export, and live video inference pipeline.
- OCR pipeline using Google Vision API and Tesseract; object identification with FLANN/SIFT.
- Serverless architecture on AWS (Lambda, API Gateway, S3).

### IQVIA — Software Engineer

Sep 2019 – May 2021

*India*

- Built a computer vision system for classifying life science documents — text analysis, spatial layout detection, and image analysis using Faster RCNN and Mask RCNN.
- Developed ML deployment infrastructure: Docker + Kubernetes + Helm templates with FastAPI microservices and GitLab CI/CD pipelines.
- Classification tracked using F1, Precision, and Recall.

India

- Contributed to CI/CD pipelines for banking applications using Jenkins and TeamCity, with automated post-deployment testing.

## EDUCATION

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### MSc (Hons) Artificial Intelligence — University of Essex, UK

Jan 2022 – Jan 2023

Dissertation: Federated self-learning chatbot with privacy-preserving dialogue adaptation. Research spans federated learning, NLP, computer vision, and medical image segmentation (HuBMAP Kaggle, top 14%).

### BEng Electronics & Communications — RGPV, India

May 2007 – May 2011

## PUBLICATIONS

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*MP-CFL: Matrix-Profile Clustered Federated Learning for Communication-Efficient Training under Data Heterogeneity*

[IEEE Transactions on Neural Networks and Learning Systems \(TNNLS\)](#) · Under Review, 2025

*Evaluating Dialogue Adaptability: A Comparative Study of Self-Feeding Mechanisms in Federated and Centralized Chatbot Architectures*

[Workshop on Secure AI Applications \(SAIA 2025\)](#) · [IEEE](#)

*Privacy-Preserving Self-Learning Chatbot with Federated Intelligence*

[International Conference on IoT, Electronics and Mechatronics \(IEMTRONICS 2025\)](#) · [Springer](#)

## TECHNICAL SKILLS

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|                           |                                                                              |
|---------------------------|------------------------------------------------------------------------------|
| <b>Languages</b>          | Python, C++, SQL                                                             |
| <b>ML Frameworks</b>      | PyTorch, TensorFlow, ONNX, Scikit-learn                                      |
| <b>Computer Vision</b>    | DINO architectures, YOLO, DeepLab, SAM 2, LoRA, PointNet, PointNet++, SPVCNN |
| <b>Federated Learning</b> | FedAvg, FedProx, TensorFlow Lite (on-device), MQTT, RabbitMQ                 |
| <b>Cloud</b>              | AWS, Google Cloud Platform (GCP), Microsoft Azure                            |
| <b>MLOps</b>              | Docker, Kubernetes, Helm, MLFlow, FastAPI, GitLab CI, Jenkins                |
| <b>Data / 3D</b>          | PDAL, GeoPandas, Shapely, NumPy, Pandas                                      |

## ACHIEVEMENTS & CERTIFICATIONS

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| Oct 2022  | Google Developer Badge — Machine Learning Bootcamp                    |
| Oct 2022  | TensorFlow Developer Certification                                    |
| Sep 2022  | HuBMAP Organ Segmentation Kaggle Competition — Top 14% globally       |
| Aug 2022  | Coursera Deep Learning Specialization                                 |
| 2018–2019 | Post Graduate Program in Artificial Intelligence and Machine Learning |

## OPEN-SOURCE CONTRIBUTIONS

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**Mindtrace** — Open-Source AI Infrastructure Framework ([github.com/Mindtrace/mindtrace](https://github.com/Mindtrace/mindtrace))

Contributing to modular micro-services for ML orchestration, observability, and agent-based automation.

**MP-CFL** — Matrix-Profile Clustered Federated Learning ([github.com/panks11/mp-cfl](https://github.com/panks11/mp-cfl))

Implementation of the MP-CFL framework — event-driven clustered federated learning with matrix-profile change detection.

**Federated Chatbot** — Privacy-Preserving Self-Learning Chatbot ([github.com/panks11/federated-chatbot](https://github.com/panks11/federated-chatbot))

Companion code for the IEMTRONICS 2025 and SAIA 2025 publications on federated dialogue learning.

**Audio File Server** ([github.com/panks11/audio\\_file\\_server](https://github.com/panks11/audio_file_server))

Lightweight server for managing and serving audio files.